

Zulqarnain Haider

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EXPERIENCE

CodeNinja Inc.

Machine Learning Engineer

Lahore, Punjab

12/2024 – 04/2026

- Designed and maintained scalable data pipelines (ELT) for ingesting and processing high-volume structured
- Developed real-time anomaly detection systems for high-volume transaction data using deep autoencoders
- Orchestrated distributed data processing and model training pipelines using Ray with GPU-aware scheduling.
- Optimized end-to-end data and inference pipelines (batching, quantization, mixed precision),reducing latency 42%
- Developed a real-time object detection pipeline using YOLOv11 for surveillance and anomaly detection tasks.
- Implemented Retrieval-Augmented Generation pipelines by integrating vector databases and embedding models, enabling context-aware inference and improving model response accuracy.

Parallel Computing Networks

AI Researcher

Islamabad,Punjab

11/2023 – 11/2024

- Designed and deployed scalable data pipelines for multimodal datasets, including ingestion.
- Implemented distributed feature extraction and embedding generation pipelines using PyTorch and Spark.
- Fine-tuned transformer-based models with PEFT and optimized inference pipelines, integrating reproducibility.
- Optimized distributed training workflows using PyTorch Distributed and Ray, reducing training times by 25%.
- Built and maintained MLOps infrastructure with Kubeflow and Airflow for automated data processing, model training, and deployment across cloud environments.

Replit

System Programmer (Rust)

San Francisco, USA (Remote)

08/2022 – 12/2023

- Engineered high-performance data processing services in Rust for handling large-scale, low-latency data flows.
- Designed and implemented asynchronous, event-driven systems using Tokio, enabling efficient handling.
- Built and optimized data access layers with PostgreSQL, including indexing strategies and query tuning to support high-throughput analytical workloads.
- Applied performance profiling and optimization techniques to data-intensive applications, achieving a 30% reduction in processing latency and improved pipeline reliability.

EDUCATION

University of Engineering and Technology (UET)

Master of Science in Computer Science

Lahore, Pakistan

01/2026 – Present

FAST National University of Computer and Emerging Sciences

Bachelor of Science in Computer Science

Islamabad, Pakistan

08/2019 – 06/2023

TECHNICAL SKILLS

Programming: Python, SQL, Rust, Bash

Machine Learning : Pytorch,JAX,kerasnlp,Deep Learning (CNNs, RNNs, LSTMs, GANs, Autoencoders), Transformer Architectures (BERT, GPT, Temporal Fusion Transformer), Time-Series Forecasting , Anomaly Detection (Autoencoders, Isolation Forest, One-Class SVM),Fine-tuning (LoRA, PEFT)

MLOps & Deployment: Docker, Kubernetes, CI/CD, FastAPI, Model Serving (TorchServe, REST APIs)